ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 1 [10 marks]

Q1(a) [4 marks]

The pixel pitch is 0.005 mm, so f_x = 16/0.005 = 3200 pixels. Sensor width = 6.4 mm. Belt field of view = 400x6.4/16 = 160 mm; spatial resolusion = 160/1280 = 0.125 mm/pixel.

Q1(b) [3 marks]

ADC step = 5/2^12 = 0.0012207 V = 1.2207 mV. Sensor sensitivity is 10 mV/N, so ΔF = 1.2207/10 = 0.12207 N. The smallest resolvable force change is approximately 0.122 N.

Calculation: $2^{12} = 4096$ levels. One step is $5/4096 = 0.0012207$ V, i.e. 1.2207 mV. Dividing by the sensitivity, $1.2207\,\text{mV}/(10\,\text{mV/N}) = 0.12207$ N. The mV units cancel, leaving N.

Q1(c) [3 marks]

$$F = 0.8187, G = 0.0906$$

Question 2 [10 marks]

Q2(a) [2 marks]

The state model is $\dot{x}_1 = x_2$, $\dot{x}_2 = (\tau - 0.05x_2 - 2.4525 \sin x_1)/0.02$. Equilibrium requires $\tau_0 = 2.4525 \sin \theta_e$, so τ_0 must be between -2.4525 and +2.4525 N m.

Q2(b) [2 marks]

Q2(c) [4 marks]

At steady state $\dot{\theta} = 0$, and I assume $\theta_{ss} = \theta_d = 90°$ because the PD controller tracks the desired angle. Thus the steady-state error is zero.

Q2(d) [2 marks]

The gravity term changes the equilibrium, so the steady-state error is still nonzero. The poles should change.

Question 4 [10 marks]

Q4(a) [2 marks]

Poles are -2±2j and Nff=8.

Q4(b) [2 marks]

With actual b=0.75 and unchanged Nff=8, k_1=2, the steady-state relation is θss = bNff/(c+bk_1) = 0.75x8/(6+0.75x2)=6/7.5=0.8. For r=1, percentage error = (1-0.8)x100%=20%.

With the actual gain, the steady-state output is $y_{ss} = b_{actual}N_{ff}r/(c+b_{actual}k_1)$ for the stated setup. Substituting the given numbers gives $y_{ss} = 0.8$ for a unit reference. Therefore $e_{ss} = 1 - 0.8 = 0.2$, i.e. a 20% error.

Q4(c) [4 marks]

The characteristic polynomial is s^3+(a+bk_2)s^2+(c+bk_1)s+bK_I. I use the target s^3+6s^2+12s+16, giving k_2=4, k_1=6, K_I=16.

Q4(d) [2 marks]